# ORTHOSES FOR SPINAL DEFORMITIES, TRAUMA, AND POST-OPERATIVE CARE

## UNIT 4

- **Orthosis**: Orthoses are external devices applied to the body to support, align, prevent, or correct deformities and improve the function of movable parts.
- **Spinal Orthoses** are specifically designed for the **vertebral column**.

## Functions of the vertebral column -

- Protect the spinal cord and its nerve roots
- Distribute axial compressive forces
- Provides axis to support the head and translates torque to axis from the limbs.

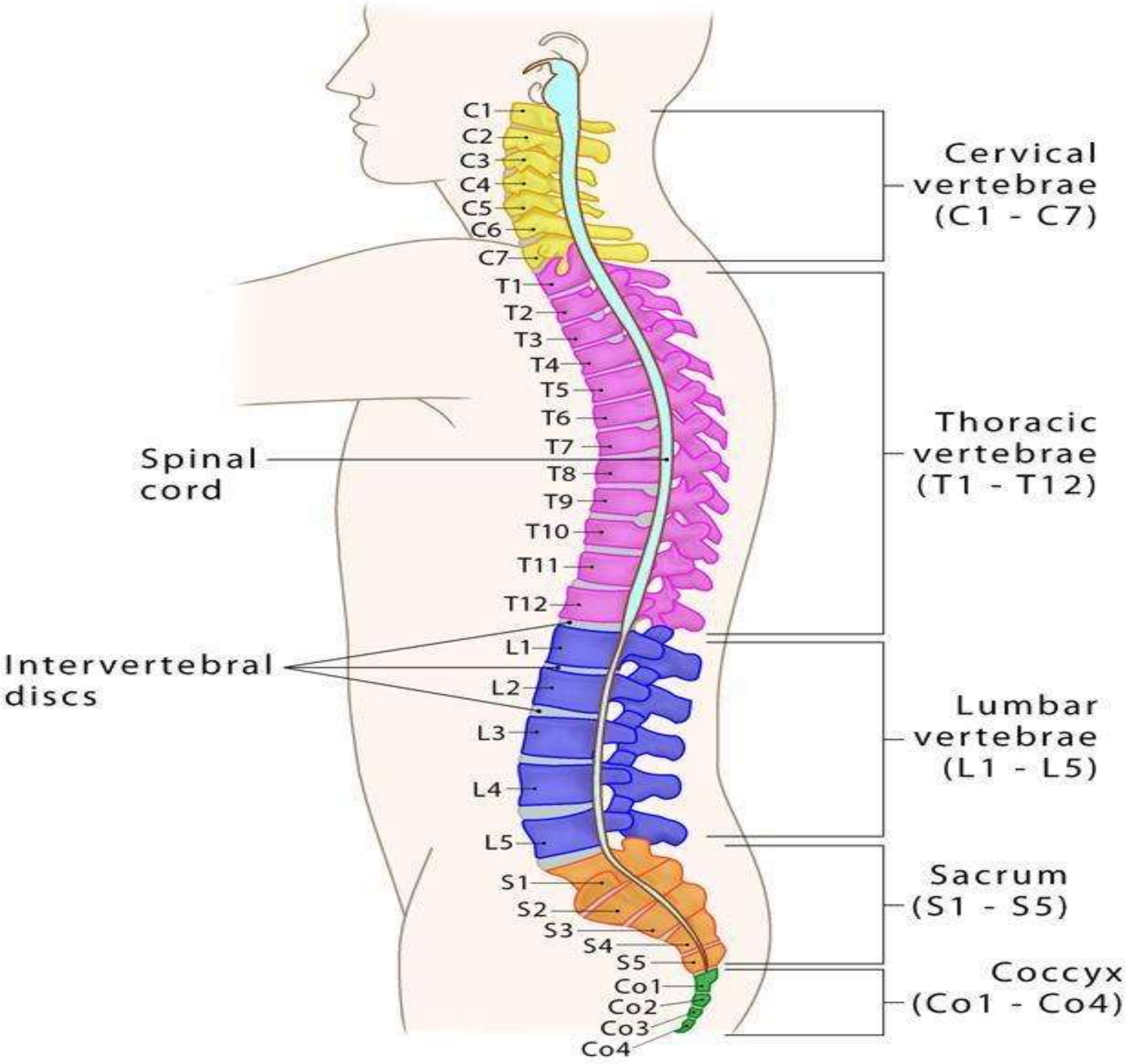
- **Clinical Need**:
  - To correct spinal deformities (e.g., scoliosis, kyphosis)
  - To immobilize after trauma (fractures, dislocations)
  - Promotion of fracture healing after spinal surgery.
  - Relief of pain by limiting motion.

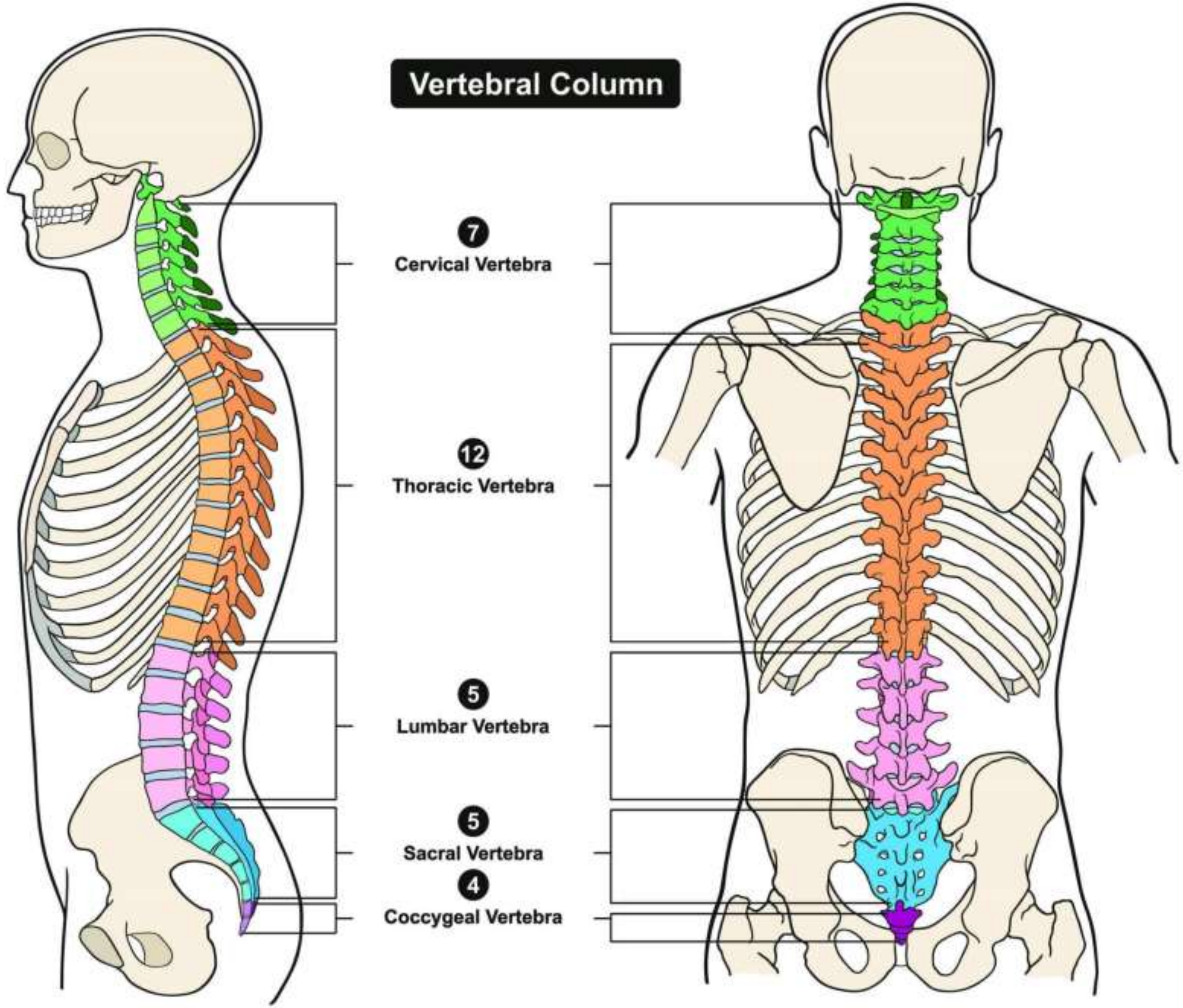
**Examples**

- Braces for scoliosis
- Cervical collars for neck injuries
- TLSO braces after spinal surgeries

# Vertebral Column

(Composed of 33 vertebrae)





Vertebral Column
7
Cervical Vertebra
12
Thoracic Vertebra
5
Lumbar Vertebra
5
Sacral Vertebra
4
Coccygeal Vertebra

# Functions of Spinal Orthoses

- **Support**: Provide mechanical support to weak or injured spinal segments.
- **Stabilization**: Prevent unwanted motion to protect healing tissues.
- **Deformity Correction**: Gradually correct abnormal spinal curves (e.g., scoliosis).
- **Pain Reduction**: Unload stressed spinal areas, improving comfort.
- **Assist Healing**: Aid bone fusion after surgery by maintaining correct alignment.

# Spinal Deformities

## 1.Scoliosis:

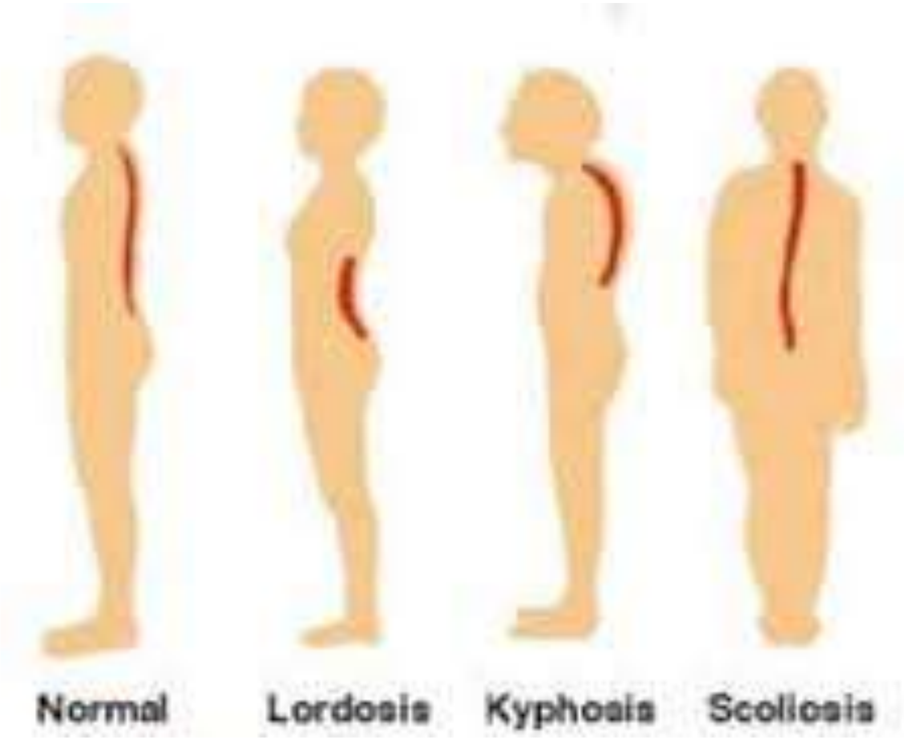
- Lateral curvature of the spine (C-shaped or S-shaped curve).

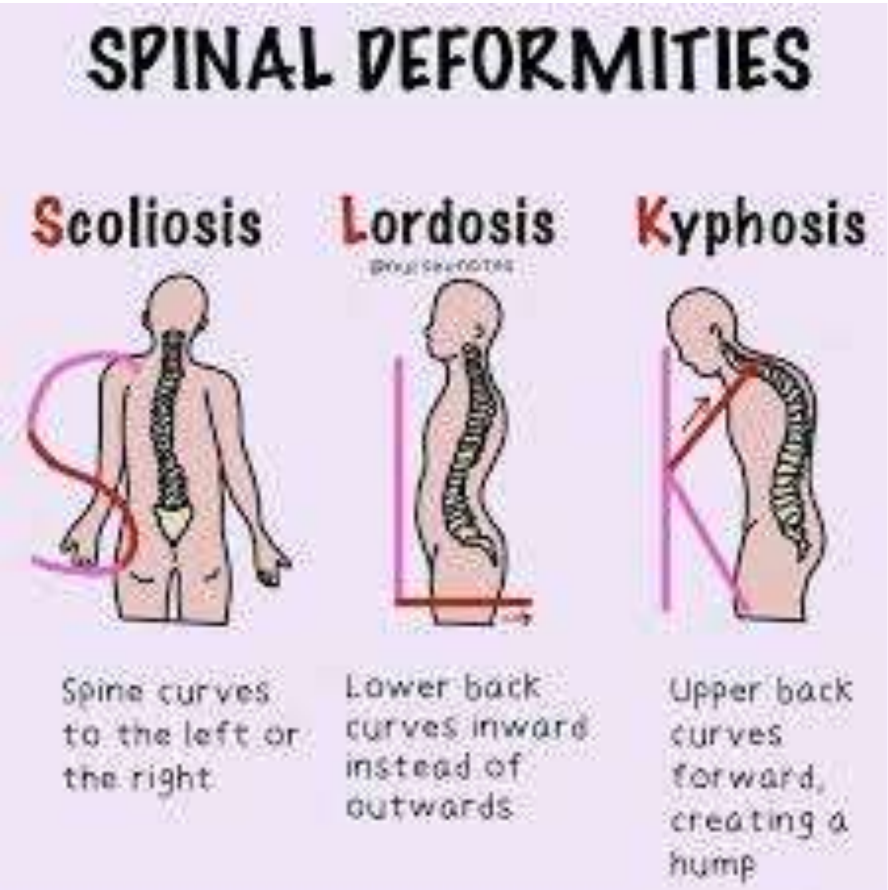
## 2.Kyphosis:

- Excessive forward rounding of the upper back (>50 degrees).
- Common in elderly (due to osteoporosis) and adolescents (Scheuermann's disease).

## 3.Lordosis:

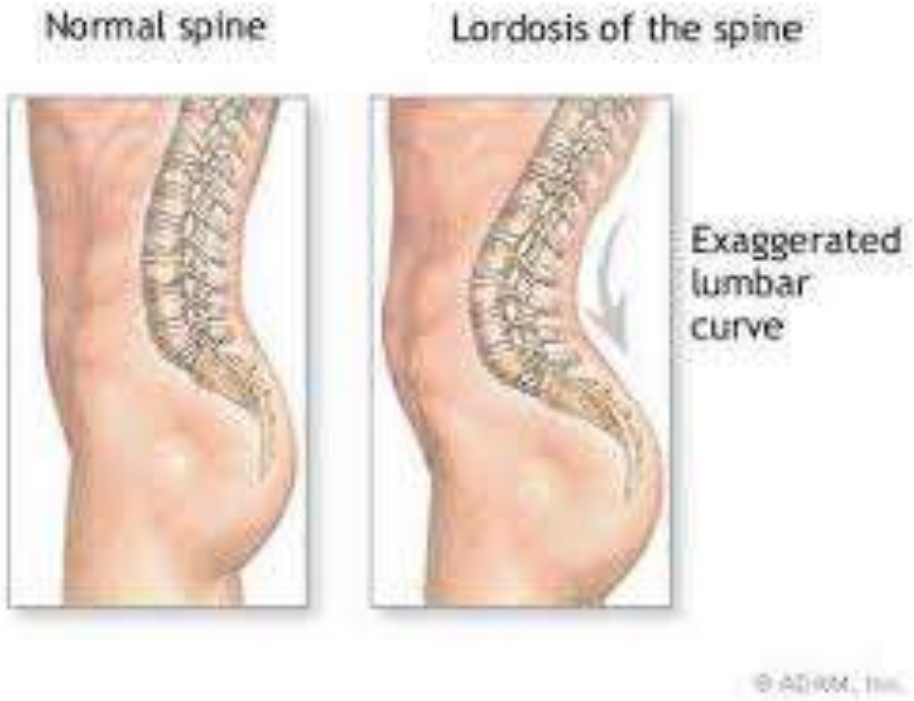
- Excessive inward curve of the lower back.
- Seen in cases of pregnancy, or muscle weakness.


Normal
Lordosis
Kyphosis
Scoliosis


SPINAL DEFORMITIES
Scoliosis
Lordosis
Kyphosis
Spine curves to the left or the right
Lower back curves inward instead of outwards
Upper back curves forward, creating a hump


Abnormal curve of spine
Ribs
Pelvis
Scoliosis


KYPHOSIS


Normal spine
Lordosis of the spine
Exaggerated lumbar curve

# Orthoses for Scoliosis and Kyphosis

### Milwaukee Brace

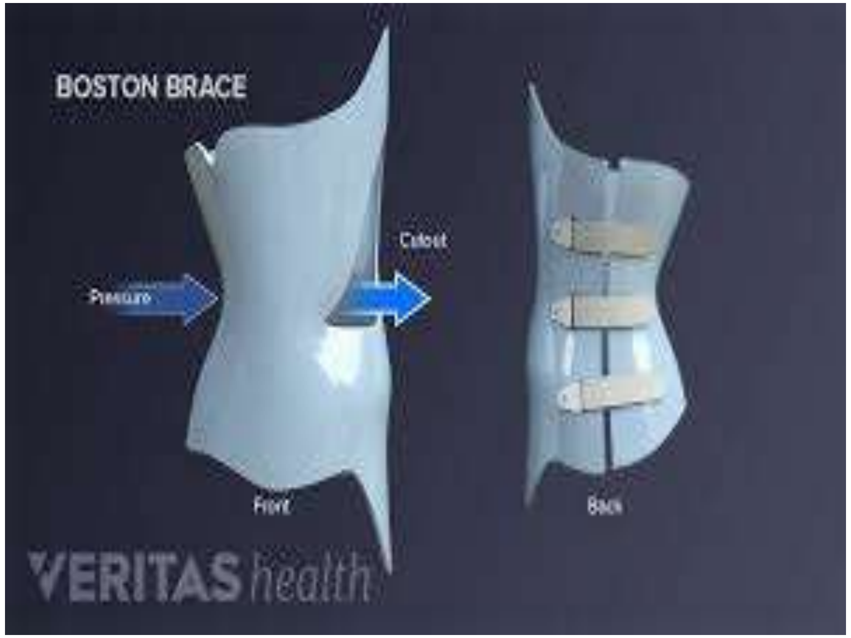
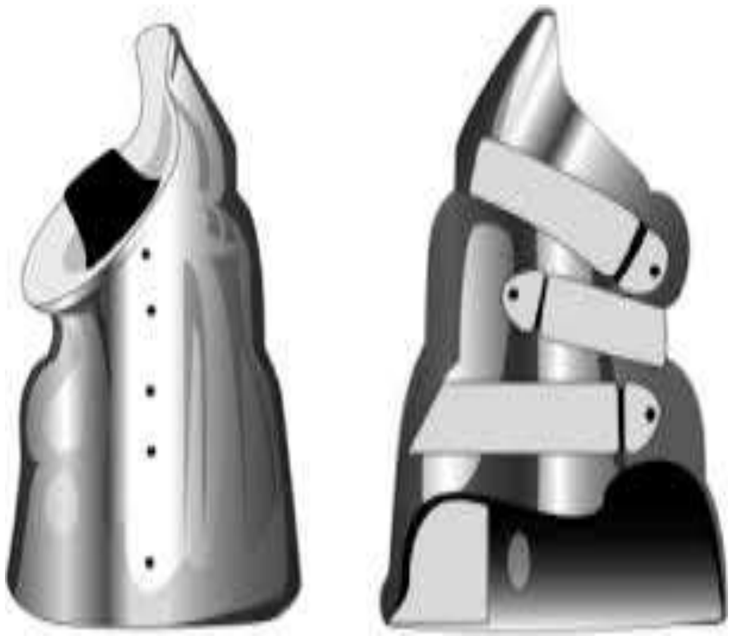
- Used mainly for scoliosis with high thoracic curves.
- Consists of a pelvic girdle, uprights, and a neck ring.
- Applies corrective forces at strategic pressure points.
- Worn up to 23 hours/day.

### Boston Brace (TLSO)

- Most common brace today.
- Underarm style; lightweight and cosmetic.
- Designed to stop curve progression rather than correct it fully.

### Charleston Bending Brace

- Only worn during sleep.
- Applies exaggerated lateral pressure to correct mild flexible curves.

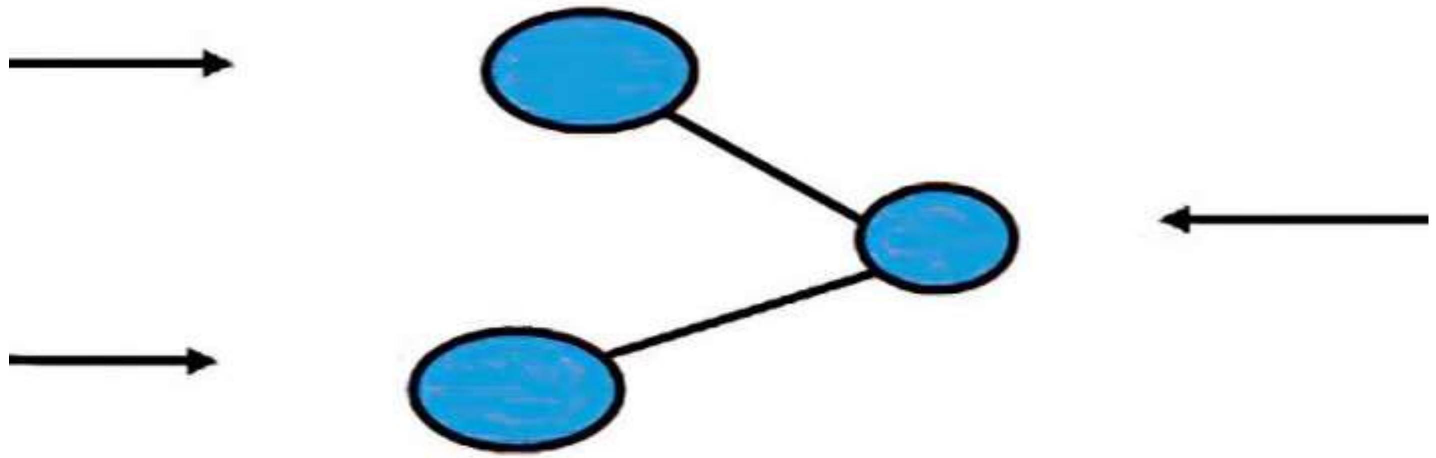

Milwaukee
Brace

BOSTON BRACE
Pressure
Cutout
Front
Back
VERITAS health


# Biomechanics of Spinal Bracing

**Three-Point Pressure System**

- Two forces in one direction, countered by a force in the opposite direction.
- Example: One pad pushes spine laterally, two counter-pads stabilize opposite sides.

- Reduces pressure on the spine while promoting corrective alignment.



# Spinal Trauma

## Types of Injuries

- **Compression Fractures**: Collapse of vertebral bodies.
- **Burst Fractures**: Shattered vertebrae, often with fragments impinging on spinal cord.
- **Dislocations**: Slipping of one vertebra over another.
- **Spinal Cord Injuries**: Damage to neural elements causing paralysis or sensory loss.

# Cervical Orthoses for Trauma

## Cervical Orthoses (CO):

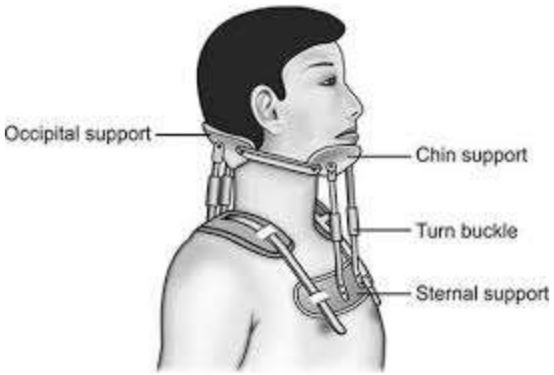
1. **Soft Collar:** Mild support for muscle strain or minor neck pain.
2. **Philadelphia Collar:** Semi-rigid support for stable fractures or post-op stabilization.
3. **Halo Vest:** Rigid fixation used for cervical spine instability or trauma.

**Philadelphia Collar**:

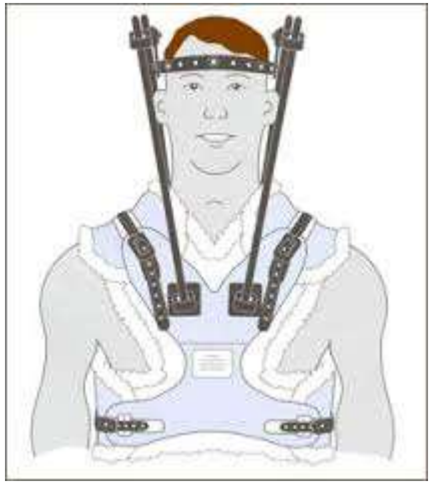
- Semi-rigid cervical collar.
- Limits neck flexion, extension, and rotation.
- Used in trauma (whiplash injuries) and post cervical spine surgery.

**Halo Vest**:

- Rigid external fixation.
- Pins attached directly to skull + vest attached to torso.
- Immobilizes entire cervical spine and upper thoracic spine.
- Allows walking while maintaining full cervical stability.


Occipital support
Chin support
Turn buckle
Sternal support


Perfect Neck Support
Effective in cases of cervical spondylitis.
Post operative care of the cervical area.



# Thoraco-Lumbo-Sacral Orthoses (TLSO)

## TLSO Brace

- Covers thoracic, lumbar, and sacral spine.
- Custom-made from thermoplastic materials.
- Used for fractures, scoliosis, and after surgeries like spinal fusion.
- Common types: Jewett brace (hyperextension), rigid body jackets.
- **Boston Brace:** Used for scoliosis; custom-made and worn under clothing.
- **Jewett Brace:** Limits flexion, used for spinal fractures.
- **Knight-Taylor Brace:** Rigid brace used for immobilization of lower thoracic and upper lumbar spine.

**Function**:

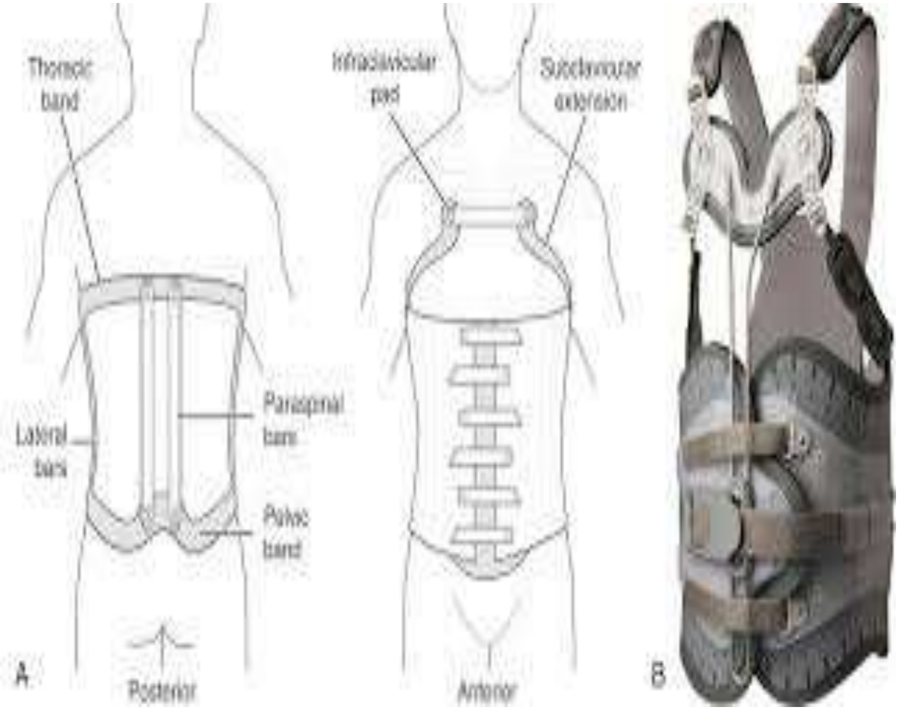
- Limits flexion, extension, lateral bending, and rotation.

## Jewett Hyperextension Orthosis

- Designed to limit forward flexion (bending).
- Used in compression fractures of the anterior vertebral body.
- Three-point pressure system:
  - Sternum pad (anterior chest)
  - Pubic pad (anterior pelvis)
  - Mid-back pad (posterior thoracic spine)

**Advantages**:

- Allows patient mobility while protecting fractured vertebrae.


Thoracic band
Lateral bars
Paraspinal bars
Pelvic band
A
Posterior
Infraclavicular pad
Subclavicular extension
Anterior
B

# Lumbo-Sacral Orthoses (LSO)

- **Chairback Brace:** Controls flexion-extension; used for lumbosacral instability.
- **Corset Brace:** Elastic support; used in low back pain and postural control.
- **Williams Brace:** Allows flexion, restricts extension; used for spondylolisthesis.

**Indications**:

- Low back pain, Lumbar fractures, Post-lumbar discectomy

**Function**:

- Restrict lumbar motion
- Decrease load on discs and vertebrae
- Promote healing

# Clinical Relevance

## In Spinal Deformities

- **Scoliosis:**
    - TLSOs like the Boston brace apply 3-point pressure to correct lateral curvature.
    - Prevent curve progression in adolescents (non-surgical cases).
- **Kyphosis:**
    - Milwaukee brace or Jewett brace provides extension force to reduce thoracic curvature.

## In Spinal Trauma

- **Fractures and Dislocations:**
  - Rigid braces (Halo vest, TLSO, LSO) immobilize the spine to allow fracture healing.
  - Prevent spinal cord damage by maintaining vertebral alignment.
- **Whiplash Injury:**
  - Soft collars reduce strain on cervical muscles.

## In Post-operative Care

- **Post Spinal Surgery (e.g., fusion, decompression):**
  - TLSO or LSO prevents excessive movement, supports healing, and improves comfort.
  - Reduces load on instrumentation and fusion sites.
- **Rehabilitation Phase:**
  - Gradual weaning from orthosis supports muscle retraining and spine realignment.

**Duration**:

- Typically worn for 6-12 weeks post-surgery

# Materials in Spinal Orthoses

## Material Choices:

- **Thermoplastics**: Moldable, strong, lightweight
- **Metals (Titanium/Aluminum)**: High strength, used in halo systems
- **Foam**: Soft interface between skin and hard materials

**Design Priorities**:

- Comfort for long-term wear
- Biocompatibility to prevent allergic reactions

# PRINCIPLES AND COMPONENTS OF UPPER LIMB ORTHOSES

Upper limb orthoses are devices designed to **support, align, prevent, or correct deformities**, or to **improve the function** of the **shoulder, arm, elbow, forearm, wrist, and hand**.

These orthoses play a crucial role in:

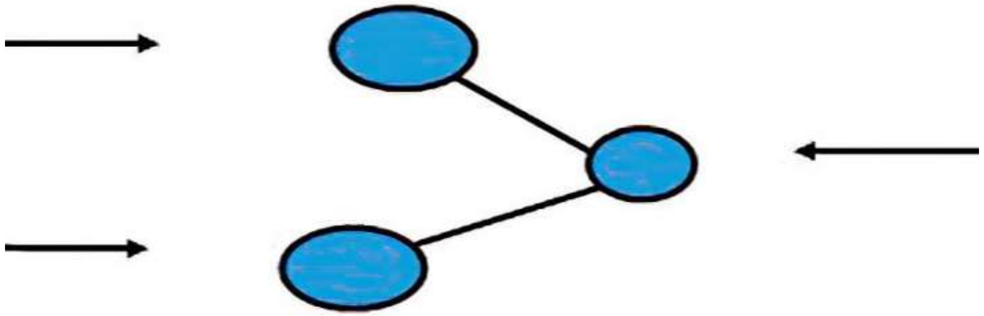
- Post-injury recovery
- Post-surgical rehabilitation
- Neurological impairments (e.g., stroke, cerebral palsy)
- Congenital limb deficiencies
- Occupational therapy

# Biomechanical Principles

These ensure that the orthosis can effectively alter motion or load in a way that benefits the user:

**a. Three-Point Pressure System**

- A fundamental concept where **three forces** are applied to control joint position or movement:
  - One corrective force in the desired direction
  - Two opposing counterforces on either side
- Used for joint alignment, control, and deformity correction.



**b. Leverage and Mechanical Advantage**

- **Longer lever arms** reduce the amount of force needed.
- Enhances effectiveness while minimizing discomfort or energy use.

**c. Load Distribution**

- Forces should be spread across **broad surface areas** to prevent skin damage or pressure ulcers.

**d. Friction and Pressure Management**

- Materials used should **reduce friction** and conform to body contours.

# Functional Principles

**Functional Principles**

Upper limb orthoses serve specific purposes, such as:

**a. Support**

- For weak or paralyzed muscles (e.g., wrist-drop splint)

**b. Protection**

- Post-operative immobilization of joints (e.g., elbow immobilizer)

**c. Assistance**

- To assist in performing daily activities (e.g., feeding orthoses)

# Patient-Centered Principles

Design should consider:

- **Comfort**: Essential for long-term use
- **Cosmesis**: Aesthetic appeal improves compliance
- **Weight**: Should be lightweight for easy handling
- **Durability**: Especially for active users
- **Ease of donning/doffing**: Should be usable independently if possible

# Classification of Upper Limb Orthoses

## By Anatomical Location

1. Shoulder orthoses (SO)
2. Elbow orthoses (EO)
3. Wrist orthoses (WO)
4. Hand orthoses (HO)
5. Wrist-Hand orthoses (WHO)
6. Elbow-Wrist-Hand orthoses (EWHO)
7. Shoulder-Elbow-Wrist-Hand orthoses (SEWHO)

# By Function

| Type | Description |
| --- | --- |
| **Static orthosis** | No moving parts. Maintains a fixed position (e.g., resting hand splint). |
| **Dynamic orthosis** | Contains springs or elastic bands to allow or resist movement. |
| **Functional orthosis** | Enhances or restores specific functional tasks (e.g., feeding aid orthosis). |
| **Assistive orthosis** | Supports movements in patients with partial motor loss. |

# Components of Upper Limb Orthoses

**A. Structural Frame/Base**

- The rigid or semi-rigid structure that provides the main support.
- May be made from:
  - Low-temperature thermoplastics (for custom molding)
  - Carbon fiber (lightweight and strong)
  - Metal alloys (for joints and high-load regions)

**B. Joints or Articulations**

- Allow or limit movement at anatomical joints
- Types:
  - Free-moving (e.g., elbow hinges)
  - Lockable
  - Adjustable ROM (Range of Motion) joints

## C. Straps and Closures

- Secure the orthosis to the limb
- Typically made of velcro or leather
- Should ensure:
  - Proper alignment
  - Comfort without skin irritation
  - Easy donning and doffing

## D. Padding and Liners

- Provide cushioning
- Prevent pressure injuries
- Absorb sweat for comfort

## E. Outriggers and Tension Mechanisms (Dynamic Orthoses)

- Elastic bands, springs, wires used to assist joint motion or provide resistance
- Commonly seen in hand orthoses for finger extension/flexion training

# Components of Upper Limb Orthoses

Upper limb orthoses consist of several key parts, tailored to the patient's anatomy and function

| Component | Function |
| --- | --- |
| **Frame or Shell** | Rigid or semi-rigid base structure; provides support. Made from thermoplastics, carbon fiber, or metal. |
| **Joints or Hinges** | Allow or restrict controlled motion (e.g., elbow hinge, wrist flexion stop). |
| **Straps and Fasteners** | Secure the orthosis to the limb. Usually velcro or buckles. |
| **Padding/Liners** | Ensure comfort, reduce pressure points, absorb sweat. |
| **Dynamic Elements** | (In functional/dynamic orthoses) include springs, rubber bands, or outriggers for assisted motion. |
| **Control Units** | (For advanced orthoses) include EMG sensors, motors, microcontrollers in powered or robotic orthoses. |

# Functional Objectives of Upper Limb Orthoses

The **primary goals** of using upper limb orthoses are:

- **Stabilization**
  - Prevent movement of injured or healing joints (e.g., post-fracture elbow orthosis).
- **Support Weak Muscles**
  - Compensate for muscle weakness due to nerve injury or stroke (e.g., wrist-hand orthosis for wrist drop).
- **Assist Movement**
  - Allow controlled range of motion during rehabilitation.
- **Prevent Deformities**
  - Used in spasticity or contractures to maintain functional alignment (e.g., resting hand splints).
- **Enhance Function**
  - Restore ability to grasp, hold, or manipulate objects (e.g., dynamic hand orthoses).

# Examples

## Wrist-Hand Orthosis (WHO)

Commonly used in **radial nerve palsy**, **stroke**, or **carpal tunnel syndrome**.

**Functional Objective**:

- Maintain wrist extension and stabilize hand posture to allow functional grasp.

**Design Features**:

- Volar splint keeps the wrist in a neutral or slightly extended position.
- May include finger slings or thumb supports.
- Lightweight and worn during daily activities.

**Example**:

- **Cock-up splint** – Prevents wrist drop and assists hand function in patients with radial nerve injury.

# Elbow Orthosis (EO)

Used after elbow fractures, ligament injuries, or to manage spasticity.

**Functional Objective:**

- Restrict or allow controlled elbow flexion/extension.
- Reduce contractures and provide support during healing.

**Design Features:**

- Hinged elbow joint with adjustable ROM stops.
- Upper and lower arm cuffs with padding.
- Allows early mobilization while protecting the joint.

**Example:**

- **Adjustable ROM elbow brace** – Used in post-operative rehab or conservative fracture treatment.

# FOOT AND KNEE ORTHOSIS

- **Lower limb orthoses** specifically support the hip, knee, ankle, and foot. It focus on improving function, stability, and mobility in patients with musculoskeletal or neurological impairments in the lower limb.
- **Foot Orthosis (FO)**
- **Knee Orthosis (KO)**

- Their design and application are based on biomechanical principles tailored to individual patient needs.
- Foot and knee orthoses **design** and **proper clinical application** greatly aid in restoring function, reducing pain, and promoting long-term musculoskeletal health.

# Foot Orthosis

- A **Foot Orthosis** is a custom or prefabricated device inserted into the shoe or worn externally to correct abnormal foot alignment, redistribute weight, alleviate pain, or enhance gait efficiency.

# Biomechanical Principles

1. **Support to Arches:** Medial longitudinal arch (most significant), lateral longitudinal, and transverse arches.
2. **Load Redistribution:** Reduces pressure on metatarsal heads or calcaneus in patients with diabetes or ulcers.
3. **Control of Motion:** Limits overpronation or supination during the stance phase.
4. **Shock Absorption:** Damping ground reaction forces during heel strike.
5. **Postural Alignment:** Influences alignment of the knee and hip through the kinetic chain.

## Design:

Foot orthoses are **custom-made** or **prefabricated** devices inserted into shoes or externally applied to the foot.

- **Types of FO Design:**
  - **Accommodative FO:** Soft and flexible materials (e.g., EVA, silicone); designed to relieve pressure and cushion deformities.
  - **Functional FO:** Rigid or semi-rigid (e.g., carbon fiber, polypropylene); controls abnormal motion and corrects alignment.
  - **UCBL Orthosis:** High-heel cup and rigid arch support; used for severe pronation control.
- **Key Components:**
  - **Shell/Baseplate** – Provides core support.
  - **Arch Supports** – Medial and/or lateral to stabilize foot arches.
  - **Posting** – Adds wedges for alignment correction.
  - **Heel Cup** – Encapsulates heel to reduce motion and improve stability.
  - **Top Cover and Padding** – Enhances comfort and pressure distribution.

# Function

- Supports foot arches and aligns bones and joints.
- Corrects foot biomechanics (e.g., overpronation, supination).
- Distributes plantar pressure evenly to avoid ulcers or calluses.
- Reduces stress on knees, hips, and lower back by improving gait mechanics.

Night orthosis

Silicon orthosis

Multifit Achilles drop foot orthosis

Footup split

Orthotic shoes

Off-the-shelf anterior shell carbon fiber ankle-foot orthosis

Push brace

Sensorimotor orthosis

Fiber orthosis

# Indications

- **Plantar fasciitis**
- **Flatfoot (pes planus)** or **high arch (pes cavus)**
- **Diabetic foot ulceration** (for offloading pressure)
- **Metatarsalgia**, **heel spurs**, **hallux valgus**
- **Post-trauma** or post-surgical support

# Rehabilitation Support

- Aids in **early ambulation** by reducing pain.
- Assists in **gait retraining** and **postural correction**.
- Prevents **recurrence of injuries**.
- Provides **long-term support** in chronic conditions like arthritis or neuropathy.

# Knee Orthosis (KO)

A **Knee Orthosis** is a supportive device that encompasses the knee joint to limit or assist movement, stabilize structures, and manage or prevent injury.

**Biomechanical Principles:**

1. **Joint Stabilization:** Controls excessive anterior-posterior or varus-valgus movement.
2. **ROM Regulation:** Hinges restrict or allow certain degrees of flexion/extension.
3. **Force Redistribution:** Particularly important in unicompartmental osteoarthritis (OA).
4. **Patellar Control:** Keeps patella aligned in trochlear groove during movement.
5. **Compression:** Reduces edema and provides proprioceptive feedback.

## Design

Knee orthoses are devices that span the knee joint to stabilize, protect, or correct alignment.

- ### Types of KO Design
  - **Prophylactic KO:** Prevents injury during high-risk sports.
  - **Functional KO:** Stabilizes injured ligaments (e.g., ACL, PCL).
  - **Rehabilitative KO:** Restricts motion post-surgery or injury to promote healing.
  - **Unloading KO:** Redistributes load in osteoarthritis by applying valgus or varus forces.
- ### Key Components
  - **Thigh and Calf Shells/Cuffs** – Anchors orthosis to the limb.
  - **Polycentric Hinges** – Allows physiological knee movement with controlled ROM.
  - **Straps and Velcro Closures** – Secure the device in place.
  - **Stops/Locks** – Control flexion-extension degrees.
  - **Padding** – Ensures comfort and prevents pressure sores.

# Function

- **Stabilizes** ligaments and joint structures.
- **Controls ROM** to prevent further injury during recovery.
- **Realigns** the knee joint (especially in varus/valgus deformities).
- **Reduces pain** by minimizing abnormal mechanical loading.

# Indications

- **Ligament injuries** (ACL, PCL, MCL, LCL)
- **Meniscal tears**
- **Post-operative care** following total knee replacement (TKR)
- **Knee osteoarthritis** (especially medial or lateral compartment OA)
- **Genu varum/valgum** (bow legs/knock knees)
- **Patellofemoral pain syndrome**

# Rehabilitation Support

- Provides **controlled mobilization** post-injury or surgery.
- Reduces **pain and swelling** by limiting harmful movements.
- Assists in **strengthening exercises** by offering safe movement range.
- Delays or prevents **surgical intervention** in chronic cases like OA.
- Enhances **confidence and balance** in patients resuming activity.

# LIMB JOINT ARTHROPLASTY

**Arthroplasty** is a surgical procedure to restore the function of a joint. A damaged joint is replaced with an artificial implant, often made of metal, plastic, or ceramic.

**Limb joint arthroplasty** is a surgical procedure in which a damaged or diseased joint in the limb (such as the hip, knee, shoulder, elbow, or ankle) is either reconstructed or completely replaced with an artificial joint (prosthesis) to restore mobility, relieve pain, and improve function.

## Objectives of Arthroplasty

- Restore joint function and range of motion.
- Relieve pain due to arthritis or trauma.
- Improve the quality of life.
- Prevent deformities and further degeneration.

# Common Joints Replaced

- Hip
- Knee
- Shoulder
- Elbow
- Ankle
- Finger joint


**Fig**-1. Partial knee replacement

2. Total knee replacement

# Types of Limb Joint Arthroplasty

| Joint Type | Type of Arthroplasty | Examples |
|---|---|---|
| Hip Joint | Total Hip Arthroplasty (THA), Hemiarthroplasty | Used in osteoarthritis or hip fracture |
| Knee Joint | Total Knee Arthroplasty (TKA), Unicompartmental Knee Arthroplasty | Common in osteoarthritis |
| Shoulder Joint | Total Shoulder Replacement, Reverse Shoulder Arthroplasty | Used in rotator cuff arthropathy |
| Elbow Joint | Total Elbow Arthroplasty | Used in rheumatoid arthritis |
| Finger/Hand Joints | Silicone joint arthroplasty | For deformities or rheumatoid damage |

# Components of a Joint Prosthesis

## A. Hip Joint Prosthesis

- **Acetabular Cup:** Fits into the pelvis.
- **Femoral Head:** Replaces the ball of the femur.
- **Femoral Stem:** Inserted into the femoral shaft.

## B. Knee Joint Prosthesis

- **Femoral Component:** Caps the end of the femur.
- **Tibial Component:** Sits on top of the tibia.
- **Patellar Button:** Replaces the surface of the kneecap.

## Materials Used:

- Titanium, Cobalt-Chromium alloys (metal parts)
- Polyethylene (for articulation surfaces)

# Types of Limb Joint Arthroplasty:

## Total Joint Arthroplasty (TJA)

- Both articular surfaces of the joint are replaced with prosthetic components.
- Common in hip and knee replacements.
- E.g., Total Hip Replacement (THR), Total Knee Replacement (TKR).

## Hemiarthroplasty

- Only one half of the joint (usually the ball component) is replaced.
- Common in hip fractures.
- E.g., Femoral head replacement only.

- **Unicompartmental Arthroplasty**
  - Replacement of only one compartment of a joint (usually the medial or lateral side of the knee).
  - Preserves more bone and allows faster recovery.
- **Resurfacing Arthroplasty**
  - Only the surface of the joint is reshaped and capped with prosthetic material.
  - Often used in younger patients to preserve bone.
- **Reverse Arthroplasty**
  - Especially used in shoulders with rotator cuff damage.
  - The ball and socket positions are reversed to improve arm lifting ability.

# Indications of Limb Joint Arthroplasty

- **Degenerative Joint Diseases**
- Osteoarthritis (most common)
- Rheumatoid arthritis
- Ankylosing spondylitis

- **Post-Traumatic Joint Damage**
- Fractures or dislocations causing long-term joint damage.

- **Joint Deformities or Congenital Defects**
- E.g., Hip dysplasia in adults.

- **Avascular Necrosis**
- Loss of blood supply leading to bone tissue death, especially in the femoral head.

- **Failed Previous Joint Surgeries**
- Revision arthroplasty may be required after implant loosening or infection.

# Post-Operative Orthotic Care

Orthotic care plays a vital role in rehabilitation and successful recovery after joint arthroplasty.

- **Joint Immobilization**
  - Temporary orthoses (braces or splints) stabilize the joint during initial healing.
  - Prevents dislocation or undue stress on the prosthesis.
- **Range of Motion (ROM) Support**
  - Controlled ROM braces help in progressive joint mobilization.
  - E.g., Hinged knee orthosis after TKR.
- **Weight Bearing Control**
  - Orthoses like walking boots or unloading knee braces control weight distribution.
  - Gradually increased as per physiotherapy protocol.

- **Gait and Posture Correction**
  - Assistive devices (e.g., crutches, walkers, shoe inserts) help improve alignment and gait during recovery.
- **Pain Reduction and Muscle Support**
  - Orthotic devices reduce pain and provide support to surrounding muscles and ligaments.
- **Prevention of Complications**
  - Supports prevent joint stiffness, deep vein thrombosis (DVT), and aid in proper healing.